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Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Personal history with meta-analysis (Meyer et al., 2002, 2012)
- Challenges to conducting comprehensive reviews
- AI to the Rescue
  - Where we are
  - Where we are going
  - What is possible
- Envisioning a Living Meta-analytic Database

Literature search method

Two complementary search strategies in every database

- **Keyword & subject-term search —** truncated organizational terms (org*, work*, employ*, job*, firm*, institution*, affect*, norm*, continu*) within three words of commit*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search —** every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only —** academic journals, books, dissertations & theses, conference papers, reports, and working papers.
- **One exception —** ERIC does not support cited-reference searching, so it received the keyword search only (13 searches in total).

Seven databases

Searched December 20, 2025 – January 7, 2026

ABI/INFORM Collection	ProQuest
Business Source Complete	EBSCO
ERIC	Ovid · keyword only
Dissertations & Theses	ProQuest
APA PsycINFO	Ovid
Scopus	Elsevier
Web of Science	Clarivate

Records identified, by database and search strategy

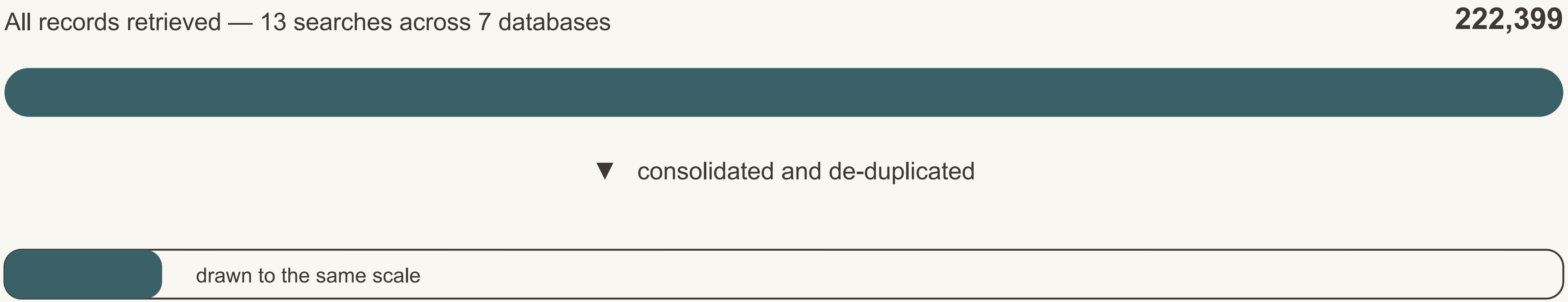
Records retrieved per database, before consolidation and de-duplication · bars share one scale

Database	Keyword search	Cited-reference search
ABI/INFORM Collection	26,659	9,642
Business Source Complete	29,551	6,382
ERIC	5,372	— not supported
Dissertations & Theses	17,936	20,410
APA PsycINFO	16,257	5,516
Scopus	25,611	19,505
Web of Science	23,898	15,660
Total	**145,284**	**77,115**

Combined: **222,399 records**

ERIC (Ovid) does not support cited-reference searching.

From 222,399 records to 22,484 unique abstracts



22,484 unique abstracts

10.1% of the initial yield — the abstracts behind every analysis that follows

How accurate is AI screening? — from prior research

Before any coding, every record has to be judged relevant or not — the first filter, deciding which studies even enter the review. In earlier published validation, the AI was compared against the expert human teams who screened by hand.

Screening — deciding which studies belong

97.2%

of relevant studies — vs **91.6%** for humans

5 meta-analyses · 30,150 abstracts screened

What the AI is being compared to

The human baseline was thorough — every record judged by hand:

2× → **1 pass**

two reviewers + a 3rd → the AI

≈ 98–99% less time — days or months of work, done in minutes to an hour.

- **Data extraction** — pulling every number out of each study — was just as strong: the AI matched expert coders at **97.0%** (humans 92.1%) across 20,476 data points.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

WHAT THE CODED ABSTRACTS REVEAL

Abstract Analysis

How the literature has grown — and what it studies.

How quickly has the literature on organizational commitment grown? Here's the annual count of articles in our collection.

Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)

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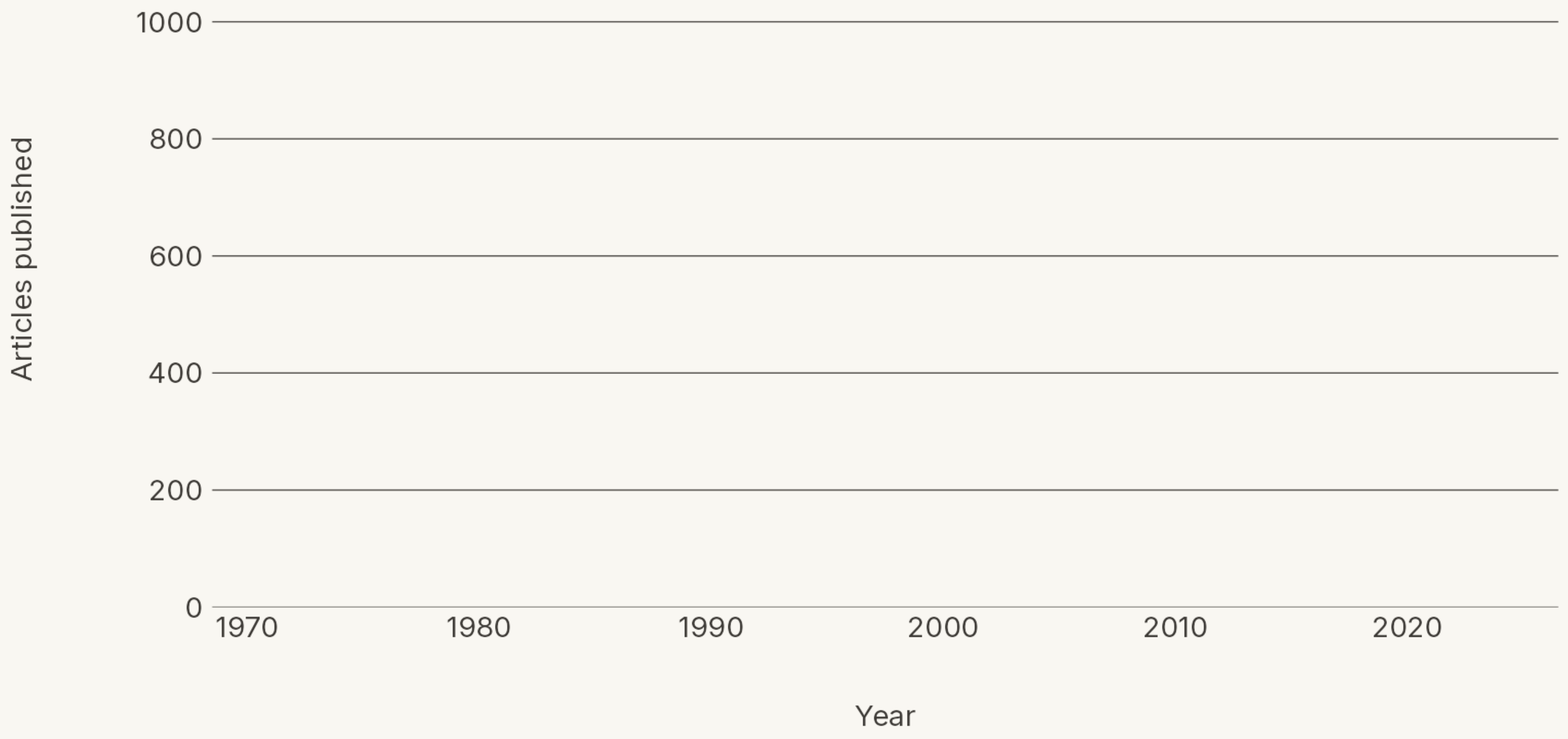

Articles Published Per Year

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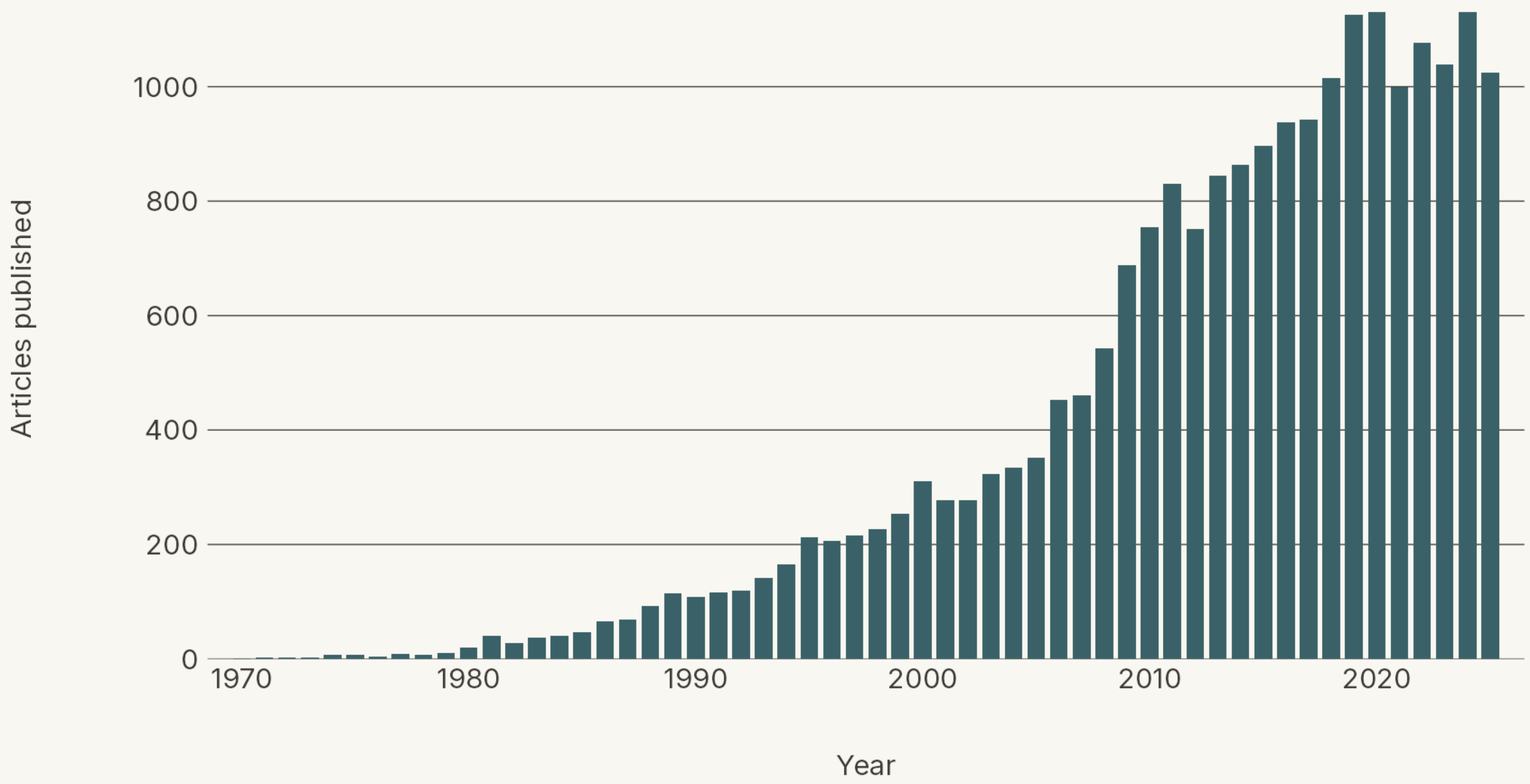

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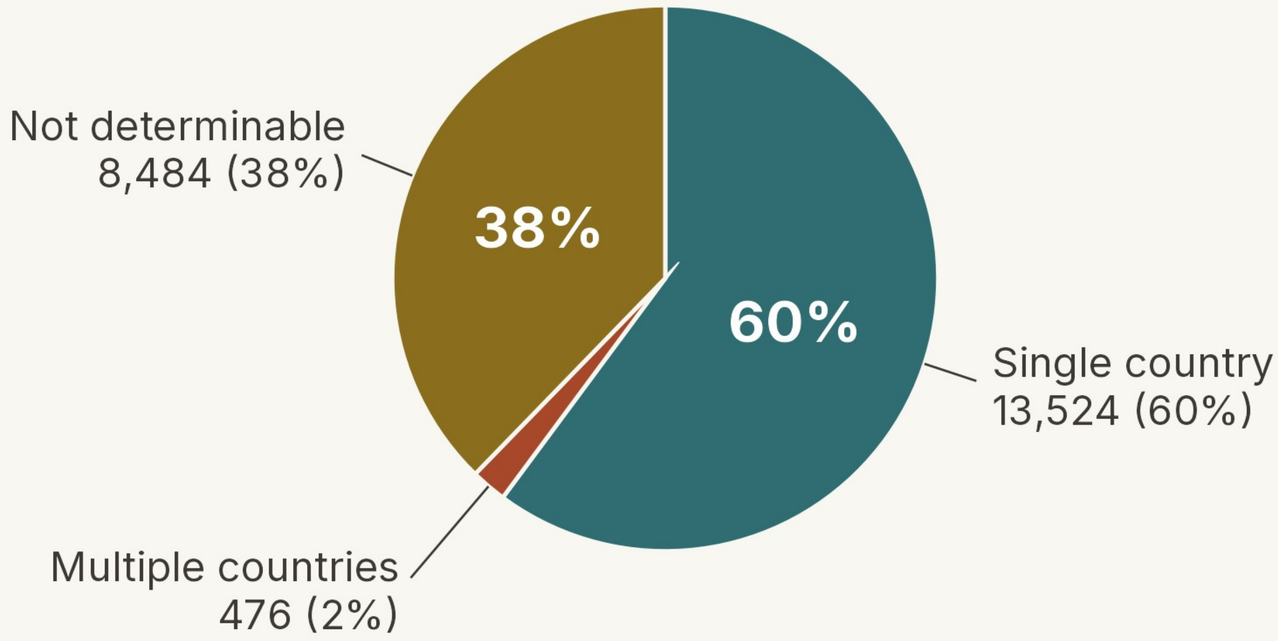
How many of the articles can we place geographically?

Country-coding status across all 22,484 unique abstracts



How country of origin was determined

Country coding criteria (each YES / NO):

- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

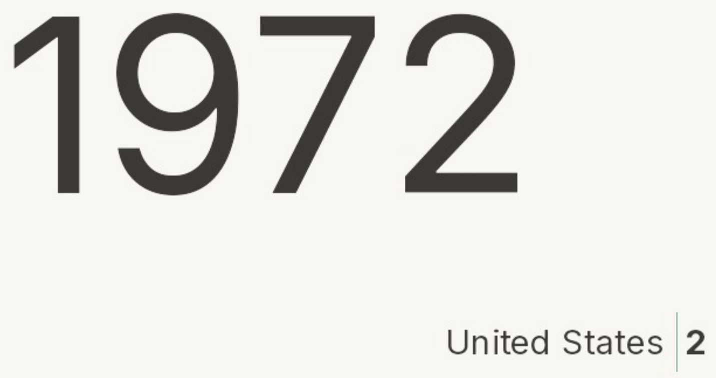
Recovering undetermined cases:

- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.

Where does this research come from? The ten countries with the most cumulative publications, ranked year by year.

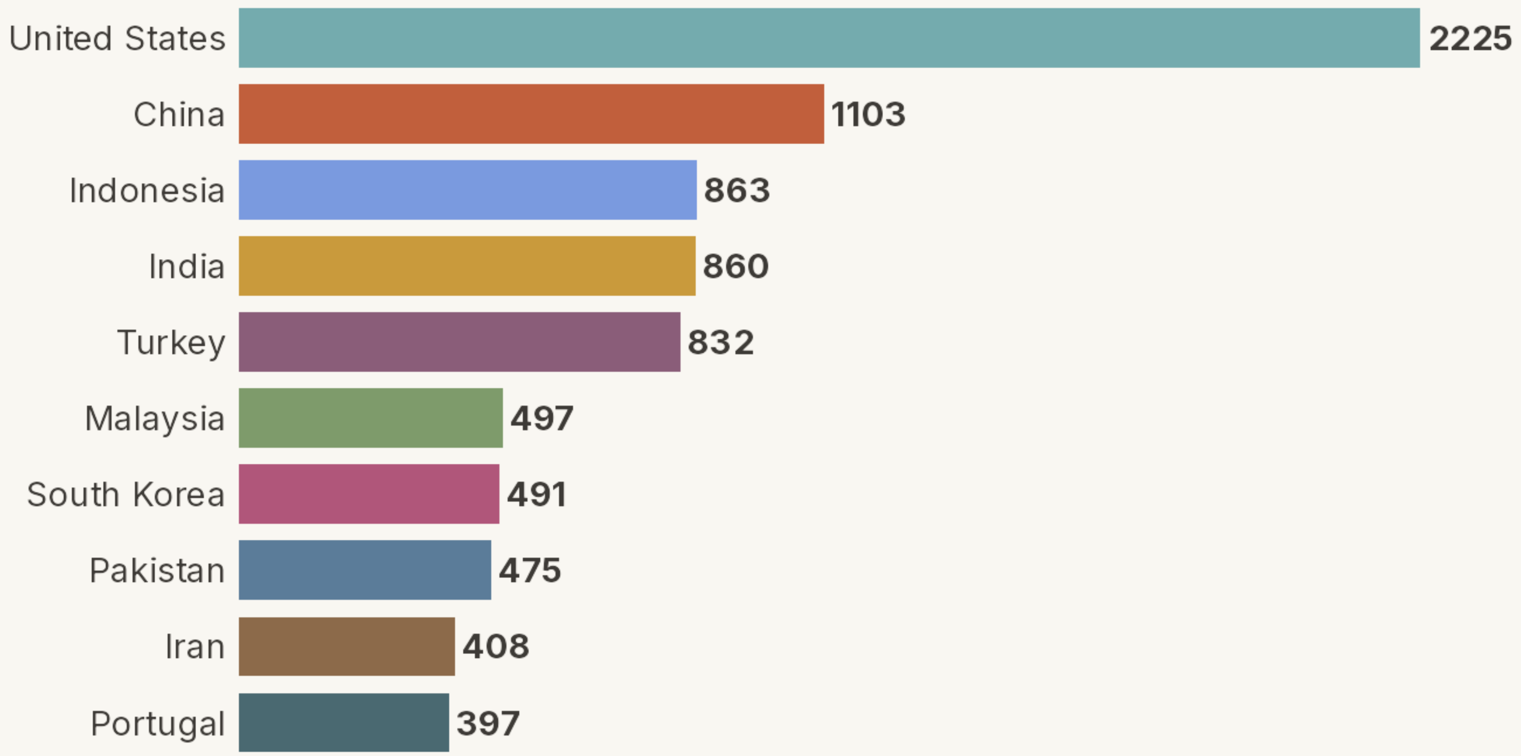
Top 10 Countries by Cumulative Publications

Top 10 Countries by Cumulative Publications



Top 10 Countries by Cumulative Publications

2025



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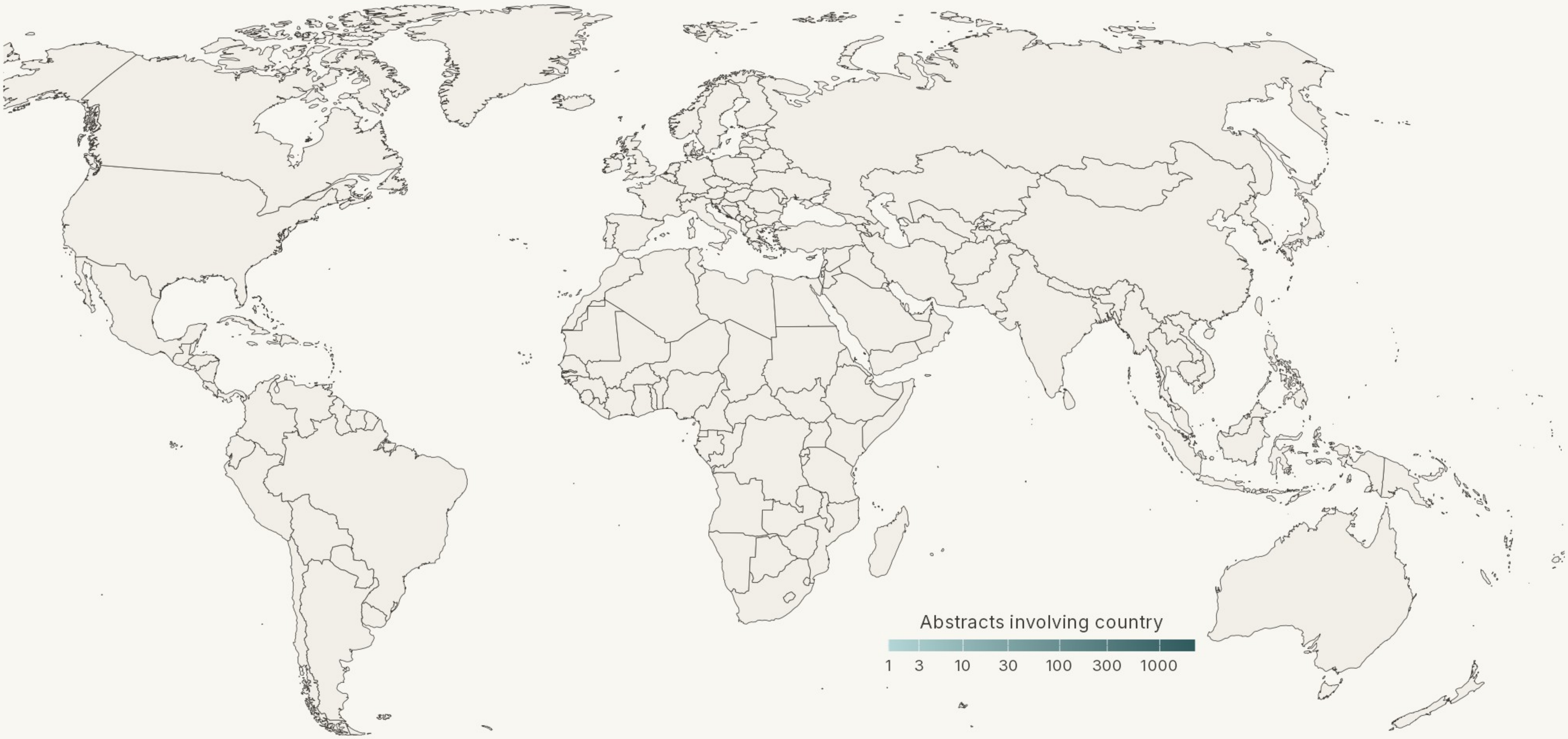
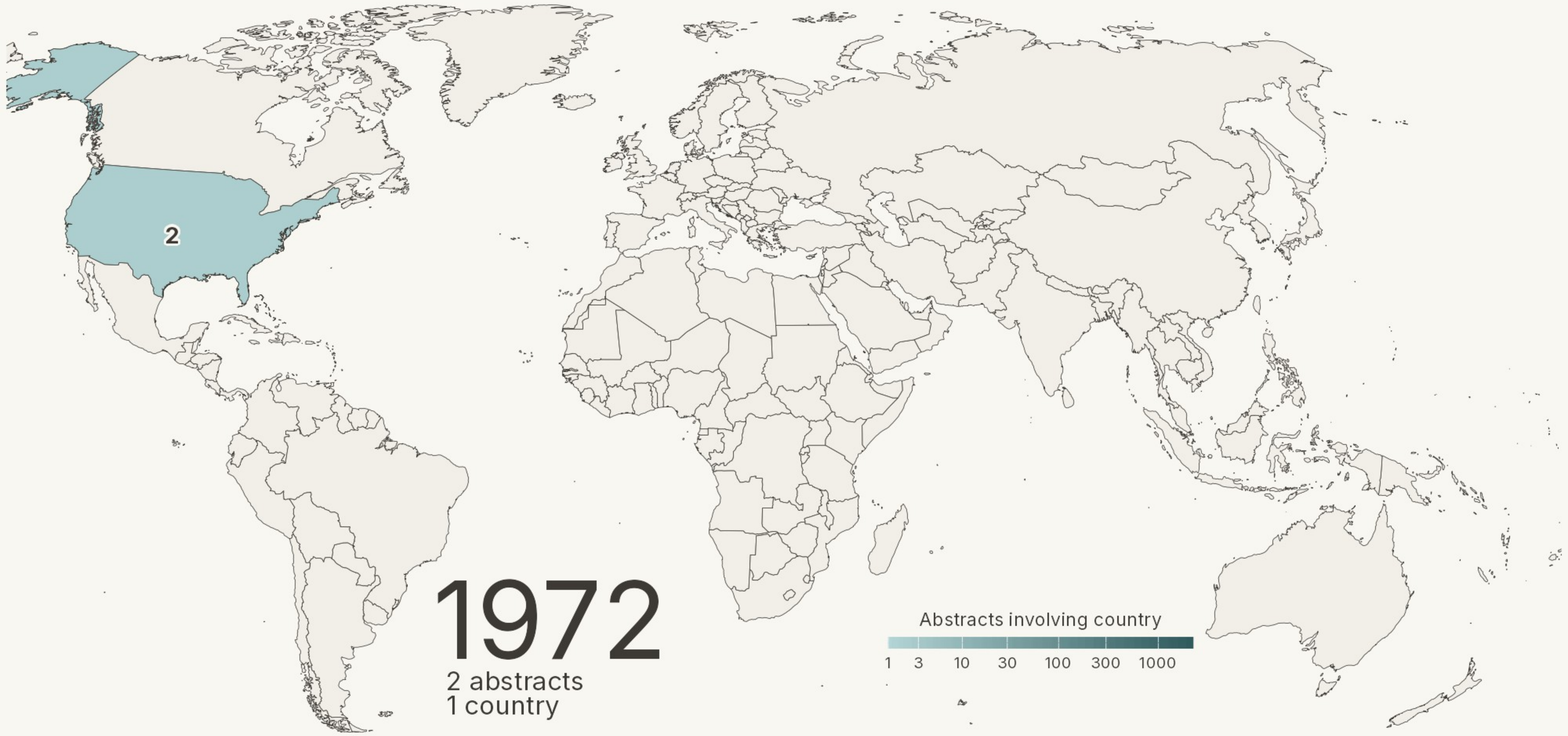
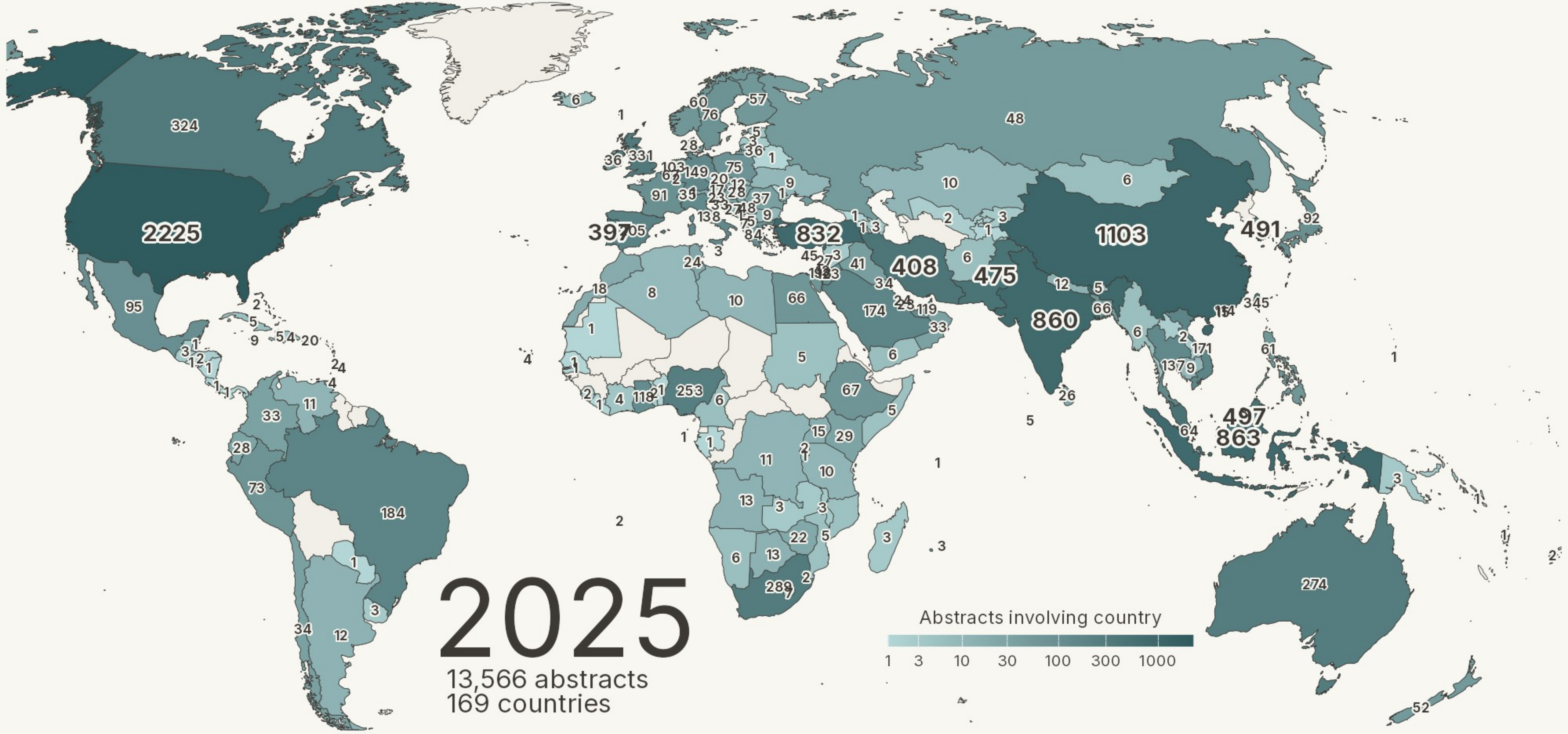
And mapped onto the globe — the geographic spread of the literature over five decades.


Abstracts involving country
1
3
10
30
100
300
1000


2
1972
2 abstracts
1 country
Abstracts involving country
1
3
10
30
100
300
1000


2025
13,566 abstracts
169 countries
Abstracts involving country
1
3
10
30
100
300
1000
2225
1103
860
863
832
497
491
475
408
397
331
324
289
274
253
184
174
149
119
118
103
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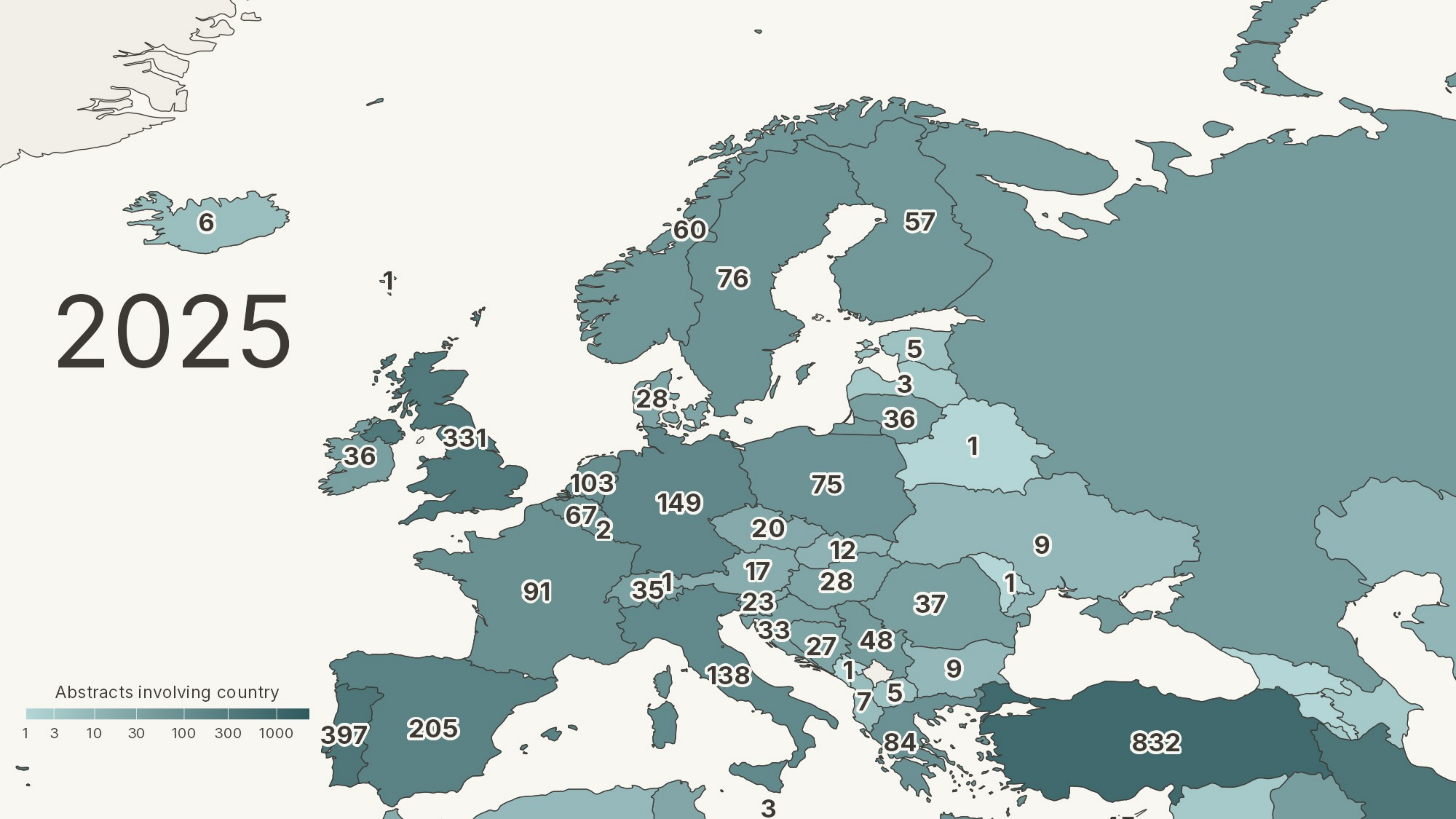

2025
Abstracts involving country
1 3 10 30 100 300 1000
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28
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103
75
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67
2
20
9
12
17
91
35
1
28
1
23
37
33
27
48
138
1
9
5
7
84
397
205
832
3

What kinds of workplaces does this literature actually study? So far we have looked at growth and geography. Now we turn to the sampled settings themselves — the industries in which these commitment studies were conducted.

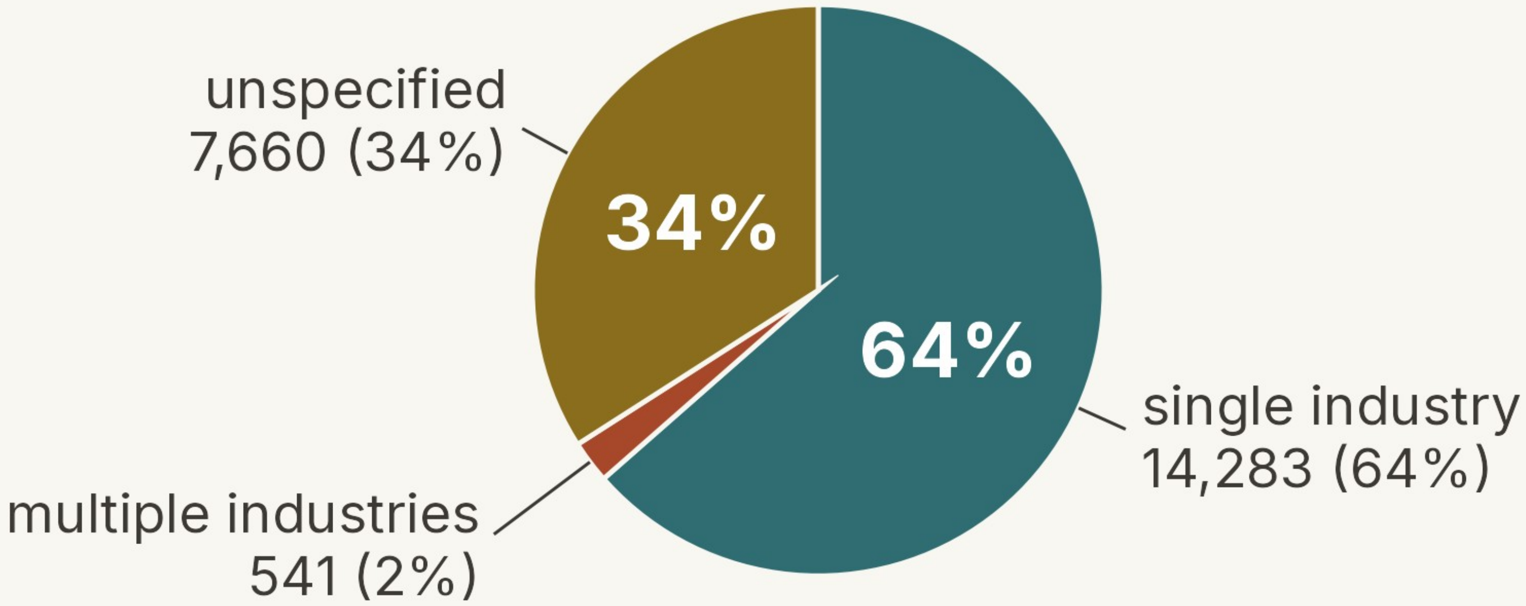
Industry taxonomy — ISIC Rev. 4

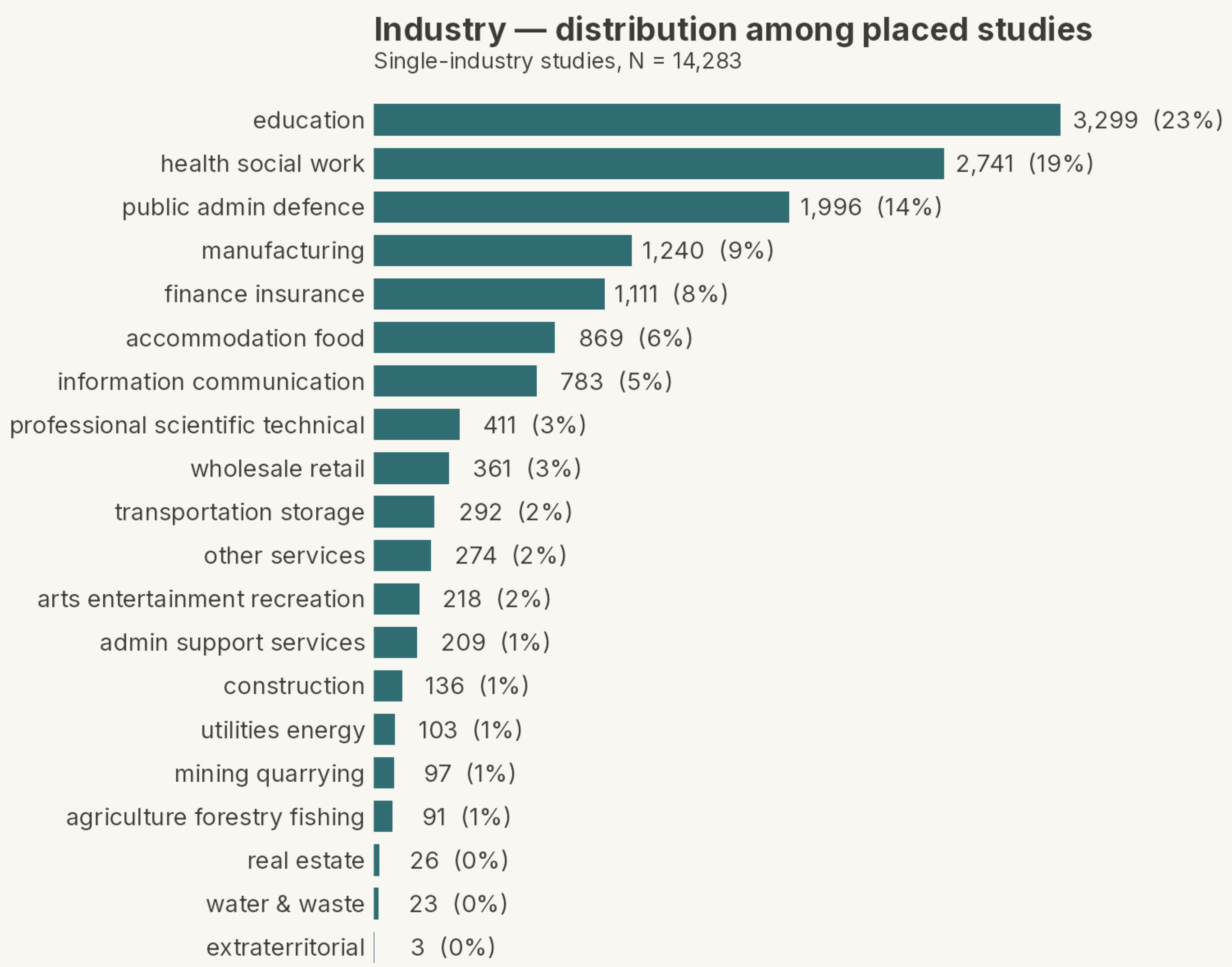
We coded each study against ISIC Rev. 4, the UN's standard industry taxonomy, which partitions the economy into 21 letter-coded sections — A (agriculture, forestry & fishing) through U (extraterritorial organizations).

Each study was tagged with the industry of its sampled workplace. Two fallbacks were used when needed: “multiple industries” for samples spanning several sections, and “unspecified” when the abstract did not name a setting at all.

Industry — coverage in three tiers

Full coded sample, N = 22,484




Industry — distribution among placed studies
Single-industry studies, N = 14,283
education
3,299 (23%)
health social work
2,741 (19%)
public admin defence
1,996 (14%)
manufacturing
1,240 (9%)
finance insurance
1,111 (8%)
accommodation food
869 (6%)
information communication
783 (5%)
professional scientific technical
411 (3%)
wholesale retail
361 (3%)
transportation storage
292 (2%)
other services
274 (2%)
arts entertainment recreation
218 (2%)
admin support services
209 (1%)
construction
136 (1%)
utilities energy
103 (1%)
mining quarrying
97 (1%)
agriculture forestry fishing
91 (1%)
real estate
26 (0%)
water & waste
23 (0%)
extraterritorial
3 (0%)

Industries told us where these commitment studies were conducted. Occupations tell us who was sampled. The two questions are distinct: a hospital can sample nurses, managers, or every employee, and each choice carries a different inferential warrant.

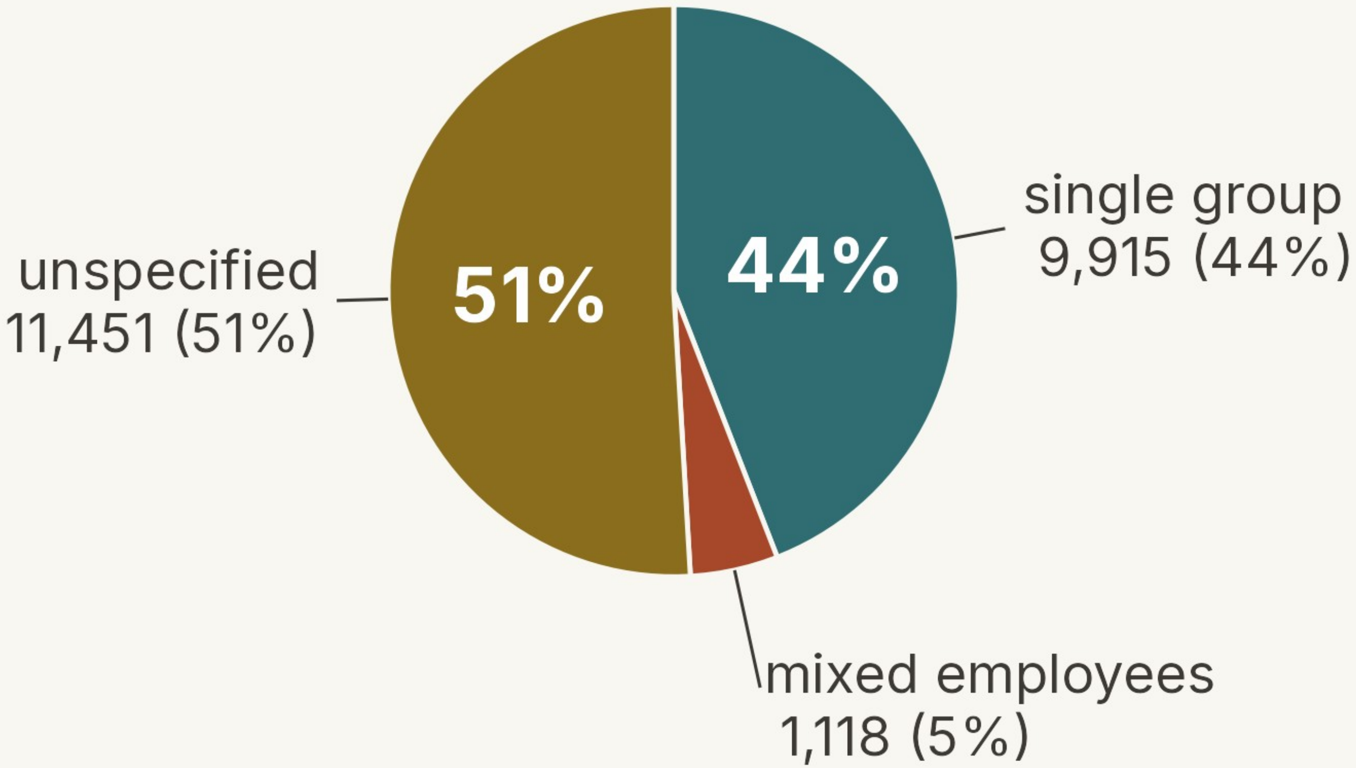
Occupation taxonomy — ISCO-08

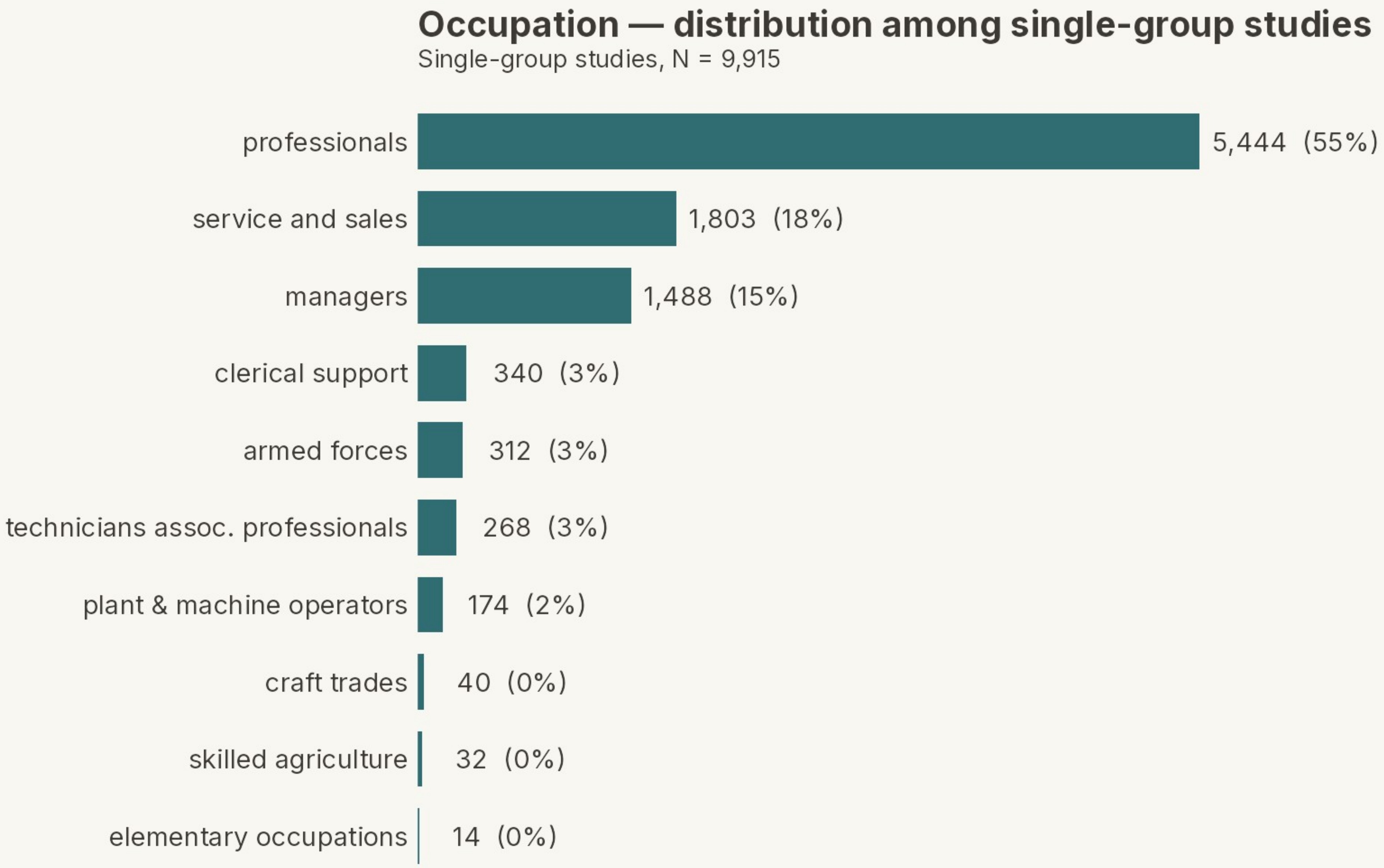
We coded each study against ISCO-08, the ILO's standard occupation taxonomy and the occupational counterpart to ISIC. At its most aggregate level ISCO sorts work into 10 major groups — 0 (armed forces) through 9 (elementary occupations) — with managers, professionals, and technicians anchoring the upper end.

As with ISIC, two fallbacks were used: “mixed employees” when the sample spanned several major groups, and “unspecified” when the abstract did not name an occupation at all.

Occupation — coverage in three tiers

Full coded sample, N = 22,484




Occupation — distribution among single-group studies
Single-group studies, N = 9,915
professionals
5,444 (55%)
service and sales
1,803 (18%)
managers
1,488 (15%)
clerical support
340 (3%)
armed forces
312 (3%)
technicians assoc. professionals
268 (3%)
plant & machine operators
174 (2%)
craft trades
40 (0%)
skilled agriculture
32 (0%)
elementary occupations
14 (0%)

Finally, how is “organizational commitment” actually used in the literature? It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

How ‘organizational commitment’ is used

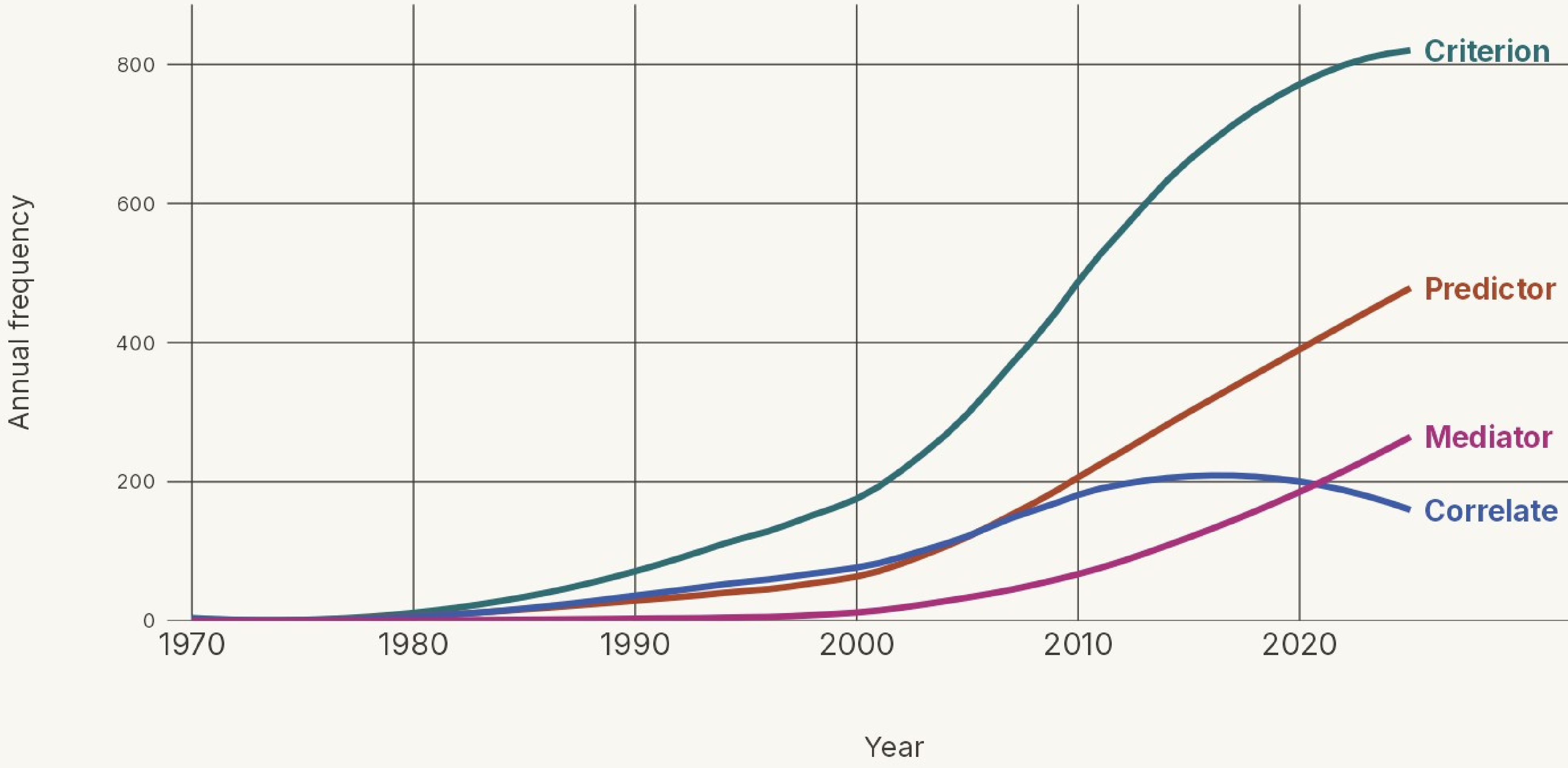
Annual count of abstracts using commitment in each role

How ‘organizational commitment’ is used

Annual count of abstracts using commitment in each role

How ‘organizational commitment’ is used

Annual count of abstracts using commitment in each role



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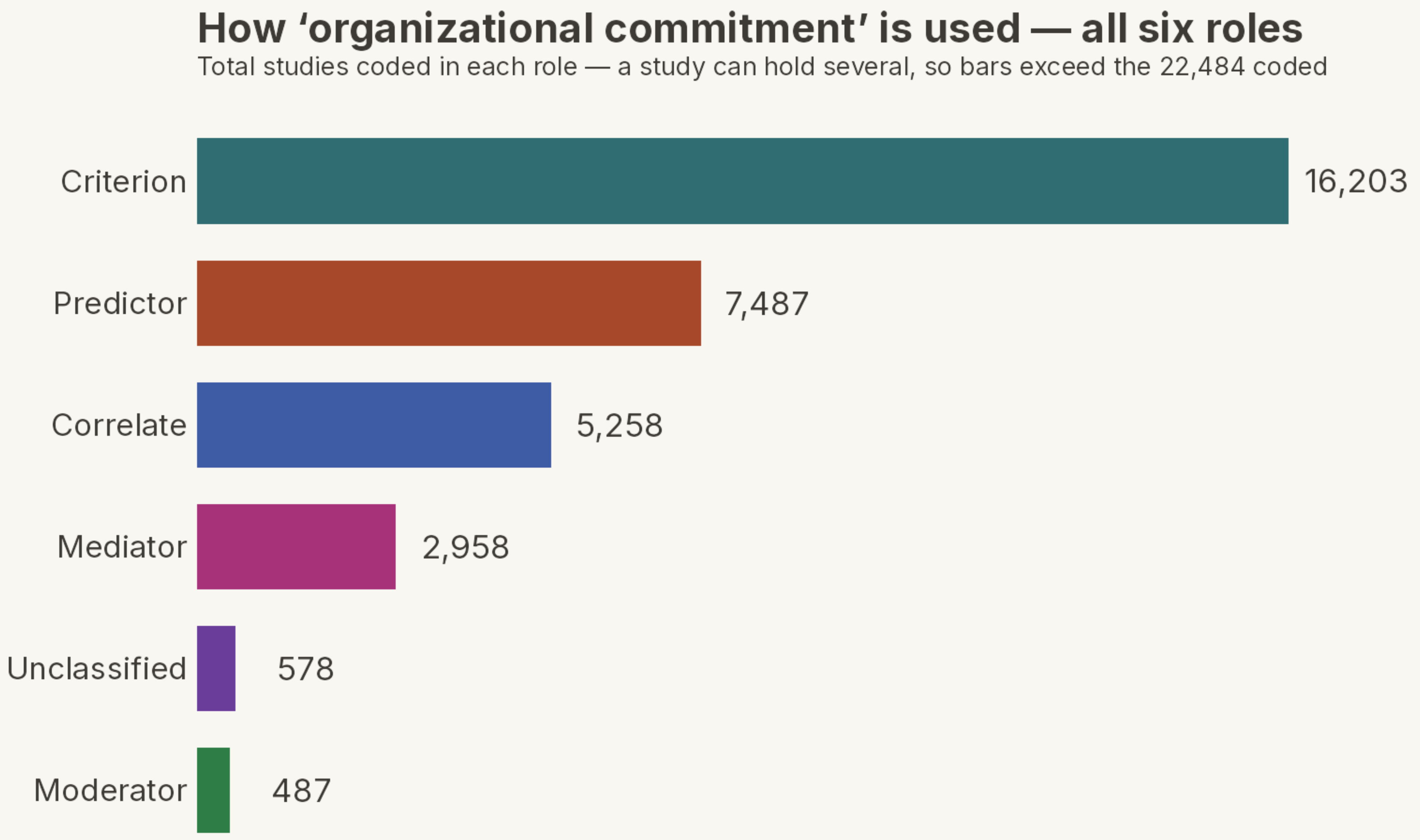

How 'organizational commitment' is used — all six roles
Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded
Criterion
16,203
Predictor
7,487
Correlate
5,258
Mediator
2,958
Unclassified
578
Moderator
487

And which specific variables show up in those roles? Across the full coded set we can look at three perspectives — what predicts commitment, what commitment predicts, and what is merely correlated with it when the study's framing doesn't assert a direction.

How did we normalize the variable labels?

Light normalization — merging only plural/spelling variants, before construct grouping

1

Extract the raw labels

Split the semicolon-separated snake_case tags coded for each study; trim whitespace and drop blanks — one label per variable mention.

2

Merge plural / spelling variants

De-pluralize (…ies → …y, a trailing s dropped) — but only when the singular is itself an already-coded label, so no merge is ever invented. The commoner spelling becomes the label.

3

Stop before meaning

No semantic or construct grouping yet — that is the later coauthor pass. Every merge is written to an audit file, so the step stays fully checkable.

Conservative by design — every merge is verified against the coded labels and logged. A cross-language unification step comes next, then the distinct-variable counts.

One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

As written across languages

job satisfaction	English
工作满意度	Chinese
satisfação no trabalho	Portuguese
iş tatmini	Turkish
satisfacción laboral	Spanish
satisfaction au travail	French
직무 만족	Korean
الرضا الوظيفي	Arabic

The unified label

job_satisfaction

one English label · **2,085** studies pooled

The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

How many distinct variables did we extract?

Light spelling normalization only — before construct grouping

Predictors of commitment

14,394

distinct variables · light model

15,838
Studies named ≥1

36,697
Total mentions

Outcomes of commitment

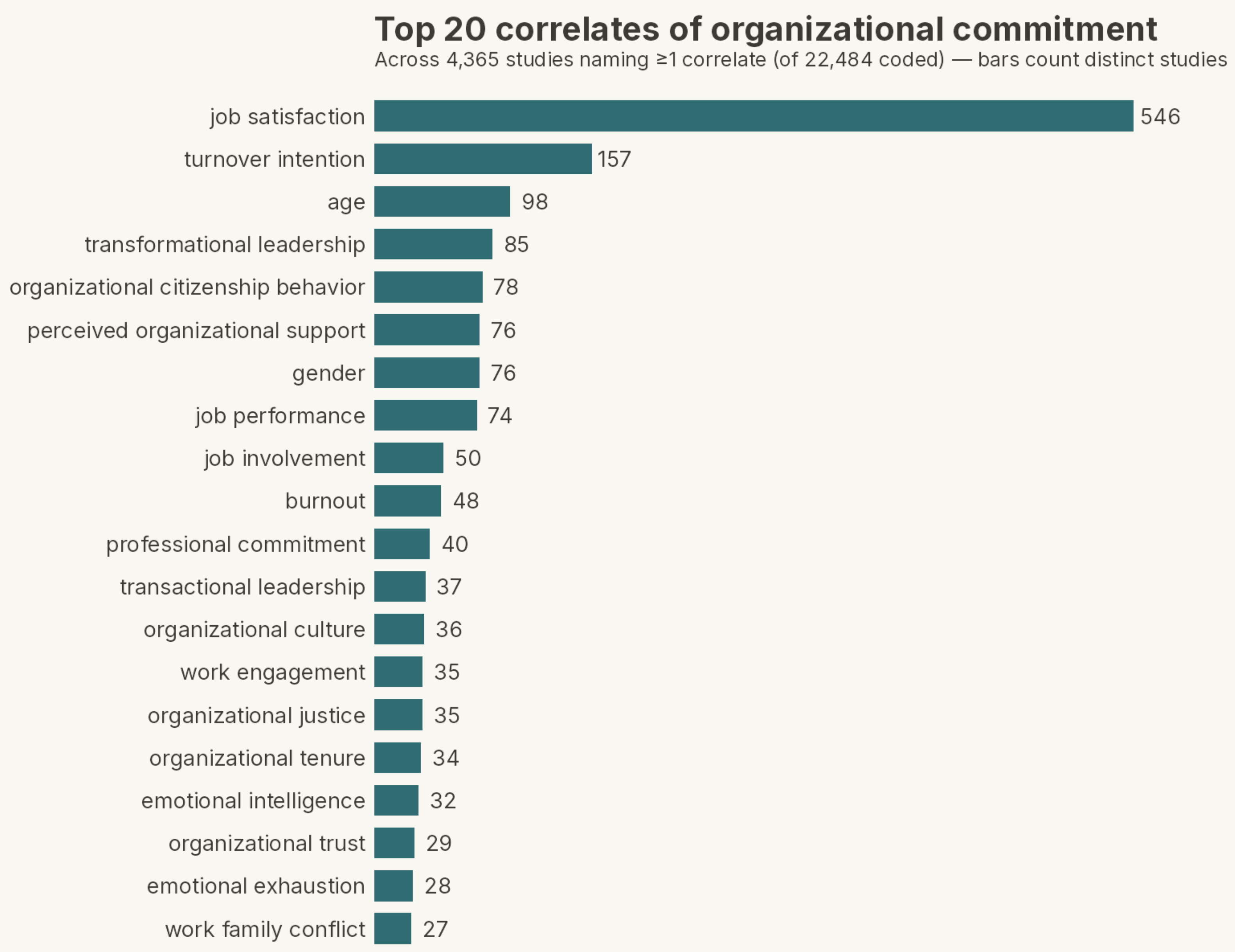
3,324

distinct variables · light model

7,462
Studies named ≥1

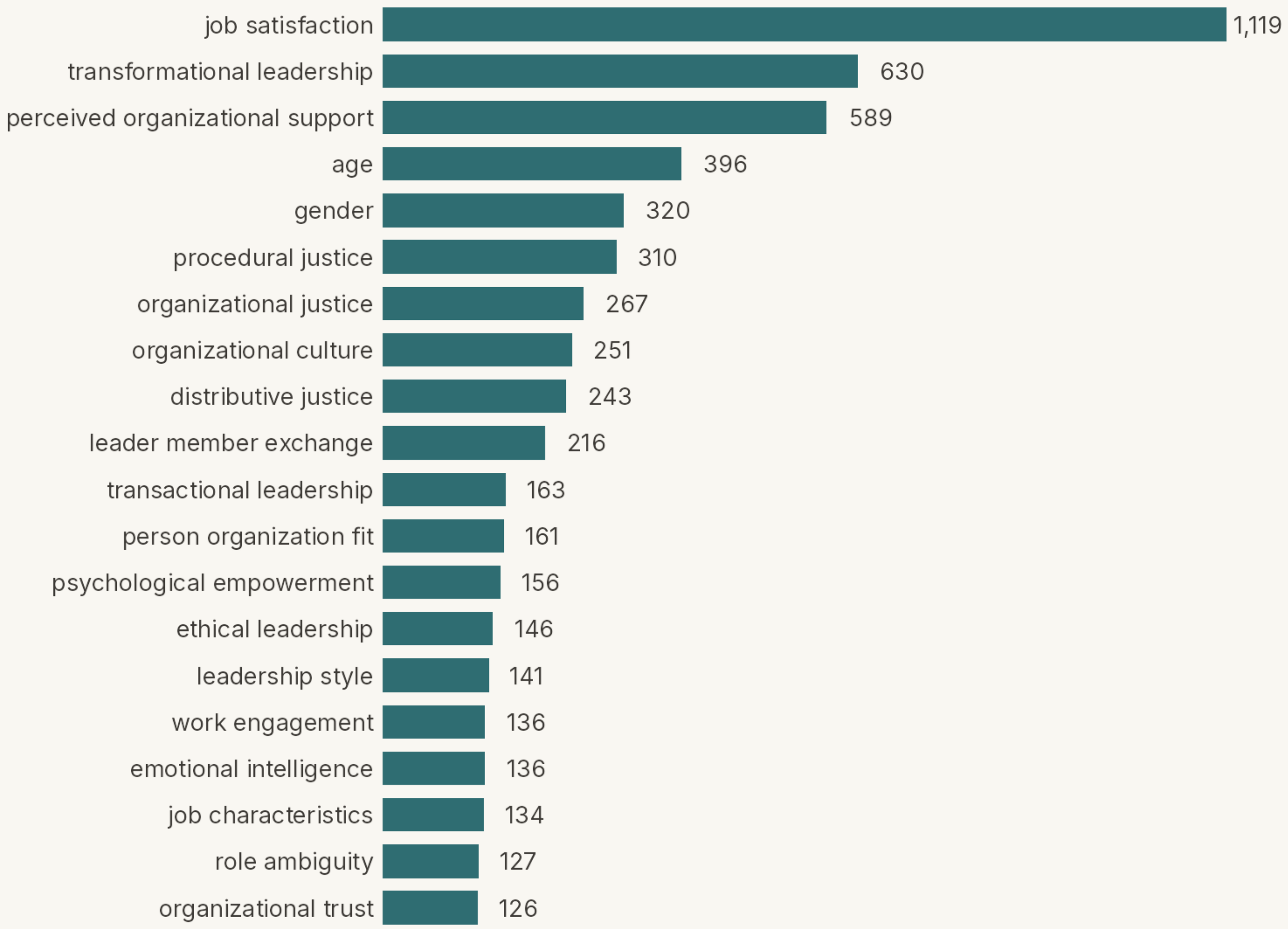
10,290
Total mentions

Light normalization is cosmetic — it merges only spellings and plurals; the real reduction comes from construct grouping.


Top 20 correlates of organizational commitment
Across 4,365 studies naming ≥1 correlate (of 22,484 coded) — bars count distinct studies
job satisfaction 546
turnover intention 157
age 98
transformational leadership 85
organizational citizenship behavior 78
perceived organizational support 76
gender 76
job performance 74
job involvement 50
burnout 48
professional commitment 40
transactional leadership 37
organizational culture 36
work engagement 35
organizational justice 35
organizational tenure 34
emotional intelligence 32
organizational trust 29
emotional exhaustion 28
work family conflict 27

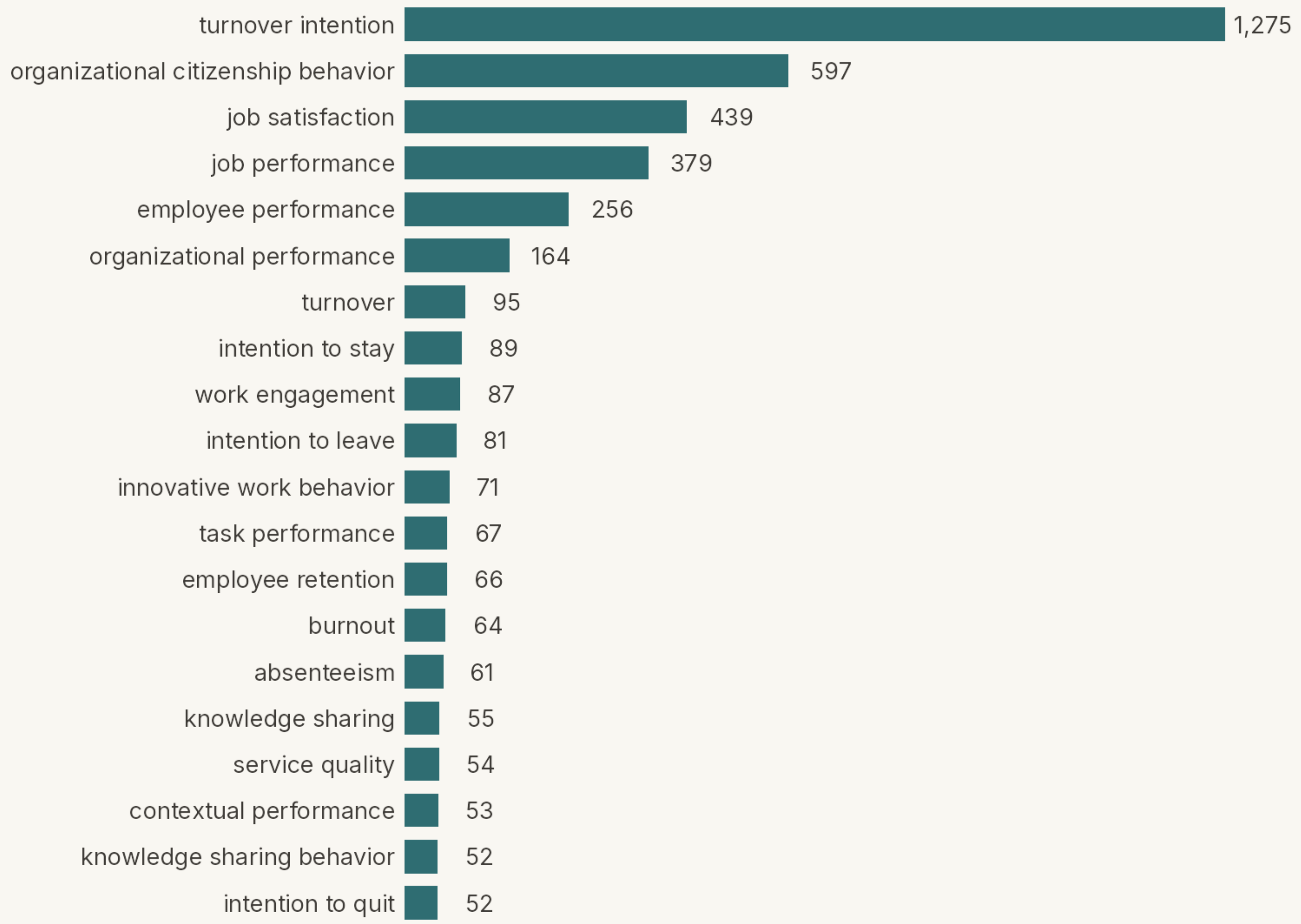
Top 20 predictors of organizational commitment

Across 15,838 studies naming ≥1 predictor (of 22,484 coded) — bars count distinct studies



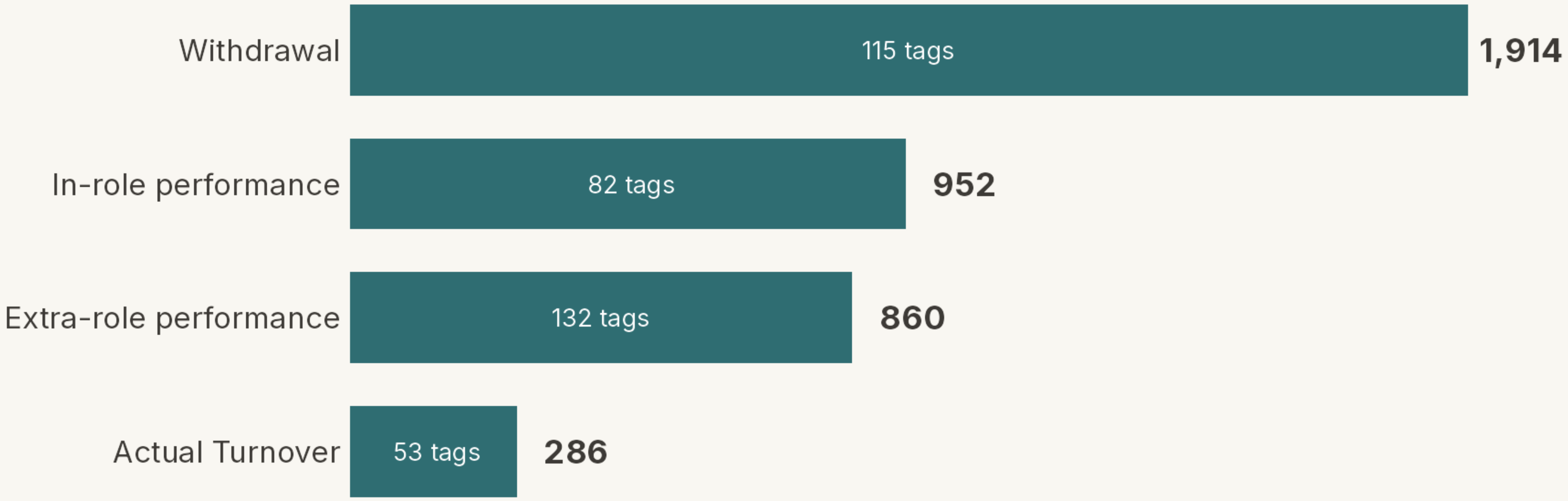
Top 20 outcomes of organizational commitment

Across 7,462 studies naming ≥1 outcome (of 22,484 coded) — bars count distinct studies



Outcomes of organizational commitment, in four families

Bar & end label = distinct studies naming ≥1 outcome in the family.
Inside each bar = number of variables (tags) collapsed into it.



FROM ABSTRACTS TO ANALYSIS

Next Steps

The work ahead, in four steps.

How accurate is AI coding? — from prior research

The labour-intensive part is the coding itself — reading each study and pulling out every sample size, mean, correlation, and reliability by hand. In earlier published validation, the AI was compared against the expert human teams who did exactly that.

Data extraction — the coding itself

97.0%

AI accuracy — vs **92.1%** for expert human coders

4 meta-analyses · 632 studies · 20,476 data points

What the AI is being compared to

One of the four meta-analyses, single-coded by hand:

130 hrs → **4 hrs**

one coder · ≈ 32× faster · the AI

12,025 numbers from 390 studies — one coder, about 20 min a study.

- **Screening** — deciding which studies belong — was just as strong: the AI caught **97.2%** of the relevant studies (humans 91.6%) across 30,150 decisions.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1. **Download the articles**
   Retrieve the full texts behind the coded abstracts.

2. **Code the articles**
   Extract effect sizes and study details from each.

3. **Formalize the variable taxonomies**
   Finalize the construct groupings from these results.

4. **Begin analyses**
   Run the meta-analytic models.

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UNIVERSITY OF GUELPH

the SYNTHESIS COMPANY of CALIFORNIA

Foundational works

The cited-reference search retrieved every record citing any of these 14 works

Allen, N. J., & Meyer, J. P. (1990). The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology, 63*(1), 1–18.

Allen, N. J., & Meyer, J. P. (1996). Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior, 49*(3), 252–276.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology, 53*(1), 39–52.

Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014). The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology, 99*(2), 222–238.

Meyer, J. P., & Allen, N. J. (1984). Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology, 69*(3), 372–378.

Meyer, J. P., & Allen, N. J. (1991). A three-component conceptualization of organizational commitment. *Human Resource Management Review, 1*(1), 61–89.

Meyer, J. P., & Allen, N. J. (1997). *Commitment in the workplace: Theory, research, and application*. Sage Publications.

Foundational works

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Meyer, J. P., Allen, N. J., & Smith, C. A. (1993). Commitment to organizations and occupations: Extension and test of a three-component conceptualization. *Journal of Applied Psychology, 78*(4), 538–551.

Meyer, J. P., Stanley, D. J., Herscovitch, L., & Topolnytsky, L. (2002). Affective, continuance, and normative commitment to the organization: A meta-analysis of antecedents, correlates, and consequences. *Journal of Vocational Behavior, 61*(1), 20–52.

Mowday, R. T., Porter, L. W., & Steers, R. M. (1982). *Employee–organization linkages: The psychology of commitment, absenteeism, and turnover*. Academic Press.

Mowday, R. T., Steers, R. M., & Porter, L. W. (1979). The measurement of organizational commitment. *Journal of Vocational Behavior, 14*(2), 224–247.

O’Reilly, C. A., III, & Chatman, J. (1986). Organizational commitment and psychological attachment: The effects of compliance, identification, and internalization on prosocial behavior. *Journal of Applied Psychology, 71*(3), 492–499.

Porter, L. W., Steers, R. M., Mowday, R. T., & Boulian, P. V. (1974). Organizational commitment, job satisfaction, and turnover among psychiatric technicians. *Journal of Applied Psychology, 59*(5), 603–609.

Powell, D. M., & Meyer, J. P. (2004). Side-bet theory and the three-component model of organizational commitment. *Journal of Vocational Behavior, 65*(1), 157–177.